Module 3 Homework 3 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:18

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

Review Later

Tickets to a movie cost $\$7$ for adults and $\$5$ for kids. Suppose 50 total people went to a showing of the movie and $\$320$ total were spent. Let x equal the number of adults and y equal the number of kids. Which of the following sets of equations matches the information given?

- $12 \times \left(x + y\right) = 320$
  $x + y = 50$
- $7x + 5y = 320$
  $x + y = 50$
- $7y + 5x = 320$
  $xy = 50$
- $\frac{320}{7} + \frac{320}{5} = 50$
  $x + y = 50$
- $\frac{320}{x} + \frac{320}{y} = 50$
  $\frac{320}{x+y} = \frac{7+5}{2}$

2

Review Later

If $3x + 7y = 420$ and $x = \frac{7}{3}y$, then what is $x + y$?

- $\frac{310}{21}$
- $\frac{1}{5}$
- $0.\overline{333}$
- 100

- 1

3

Review Later

When the decimal point of a positive number x is moved 2 places to the right, the new number is 9 times the **reciprocal** of the original number. Find the expression that accurately matches this.

- $100x = \frac{9}{x}$
- $100x = 9x$
- $\frac{x}{100} = 9x$
- $\frac{x}{100} = \frac{9}{x}$
- $100x = \frac{x}{9}$

4

Review Later

In the quadratic equation $x^2 + ax + b = 0$, where a and b are real numbers, which of the following correctly describes the relationship between a and b and the roots of $x^2 + ax + b = 0$?

- $a = (-1) \times \left(\text{sum of the roots}\right)$
  $b = \text{product of the roots}$
- $a = \left(\text{sum of the roots}\right)$
  $b = (-1) \times \text{product of the roots}$
- $a = \text{product of the roots}$
  $b = -1 \times \text{sum of the roots}$
- $a = -1 \times \text{product of the roots}$
  $b = \text{sum of the roots}$
- $a = \text{sum of the roots}$
  $b = \text{product of the roots}$

5

Review Later

Robert has fish, chickens, and dogs. There are 28 eyes total and 16 legs total. Also, there are twice as many fish as chickens. How many dogs are there?

6

Review Later

When the decimal point of a positive number x is moved 2 places to the right, the new number is 9 times the **reciprocal** of the original number. Find the value of x. Your answer should be a fraction in simplest form without spaces, e.g. 1/2

7

Review Later

Ann realizes that the price of a skirt is 80% of her money and the price of a blouse is 60% of her money. After her grandma gives her $\$16$, Ann will be able to buy both. How much money in dollars does Ann have in the beginning (before her grandma gives her money)? Your answer should be the number only, without the $\$$ sign.

8

Review Later

Find which of the following set of equations has a unique solution; i.e. each variable has a solvable value:

$x + 2y + \frac{1}{3}z = 3 + y^2$

$z - x = 4$

$x + y = 4$

$y + z = 3$

$xyz = 1$

$3x + 4y + z = 20$

$y - z + 2x = 3$

$\frac{1}{2}z + \frac{3}{4}y = 10 - x$

$z = \sqrt{-x^2}$

$y = \sqrt{-z^2}$

9

Review Later

The sum of two numbers is 76, and the sum of 25% of the first number and 15% of the second number is 14. Find the difference between these two numbers.

10

Review Later

If you plug the *root* of the quadratic equation $x^2 + ax + b$ into itself, i.e. substituting the root for the value of x, the value of the expression becomes

-1

1

$a + b$

0

- [ ] $1 + a + b$

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

The sum of the squares of two positive numbers is 800. Their product is 112. What is the sum of these two positive numbers?

12

Review Later

James and John are swimming. James swims $2x - 4$ miles in $5x - 6$ hours. John swims $10 - 2x$ miles in $x + 3$ hours. If they are swimming at the same speed, then find the value of x.

13

Review Later

If p and q are the two solutions to $x^2 + 13x - 39$, find $(p - q)^2$.

14

Review Later

A picture with dimensions 3 inches by 4 inches is placed in a picture frame of uniform width around all sides, enlarging its area to 42 square inches. What is the width of the picture frame? Your answer should be a decimal rounded to the nearest tenth, without units.

15

Review Later

Suppose $f(x) = x^2 + 17x - 39$ and $g(x) = 4x + 1$. What is the sum of all possible values of x such that $g(f(x)) = 0$?

16

Review Later

What is the sum of all integers p such that $x^2 + px + 20$ can be factored into $(x - a)(x - b)$ where a and b are integers?

17

Review Later

Suppose that $x - c$ is a factor of $x^2 + 2cx - 3$. Find the larger of the two values of c.

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

The number of owls varies inversely with the number of mice, and the number that you get from multiplying the number of owls with the number of mice is always the same. When there are $7x-5$ owls, there are $4x+8$ mice. When there are $6x-6$ owls, there are $11x+2$ mice. What is the product of the number of owls and the number of mice?

19

Review Later

Solve for x in the equation

$$\left(\frac{5-x}{4-x}\right)-\left(\frac{6-x}{5-x}\right)=\left(\frac{7-x}{6-x}\right)-\left(\frac{8-x}{7-x}\right)$$

Your answer should be a decimal rounded to the nearest tenth.

20

Review Later

Let $f_1=\frac{x+1}{x-1}$. Let f_2 be formed by substituting $\frac{x+1}{x-1}$ for x in f_1, f_3 be formed by substituting $\frac{x+1}{x-1}$ for x in f_2, and so on and so forth. Find f_x for $x=2019$.

Your answer should be a fraction in simplest form without spaces, e.g. 1/2

View Marked Questions

Finished

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